KING SAUD UNIVERSITY COLLEGE OF COMPUTER AND INFORMATION SCIENCES Computer Science Department	
CSC 111: Computer Programming - I	**Project:** Phase#1

Parking Slot Reservation System

Project Description

This project involves developing a Parking Slot Reservation System that simulates a self-service parking system in a city parking facility. The system allows drivers to select an available parking slot and choose a parking duration from predefined options.

You are required to develop a user-friendly Java program with the following features:

1. Predefined Parking Slots and Durations

The parking facility provides predefined parking durations and parking slots. Five parking durations are available: 1 Hour, 2 Hours, 3 Hours, 4 Hours, and a Day Pass. The parking price is calculated based on a fixed hourly rate, which is 5 SAR per hour. The price for a Day Pass is fixed at 30 SAR.

The system also maintains a fixed number of five predefined parking slots, each identified by a slot ID and an availability status (available or occupied). These parking options should be pre-defined and not require any user input.

2. User Interface (Main Menu)

When the program starts, a welcome message is displayed followed by the Main Menu:

- Customer (Driver)
- Parking Administrator (Store Owner)
- Exit

If the user selects **Customer**, the Customer Self-Service Menu described in part 3 is displayed. If the user selects **Parking Administrator**, the Administrator Menu described in part 4 is displayed. The user can terminate the program at any time by choosing the **Exit** option.

3. Customer Self-Service Menu

Customers can access a menu to view available parking slots and manage their reservations. The menu allows them to:

- Select Parking Slot and Duration
- Checkout
- Cancel and return to Main Menu

Customers first select an available slot and choose a parking duration. They can then proceed to checkout or cancel the transaction. Upon checkout, the system generates a formatted invoice showing the selected parking slot, parking duration, and total cost. The customer can then confirm or cancel the transaction. If confirmed, the selected parking slot is marked as occupied, and the system returns to the main menu. If cancelled, no changes are made, and the program returns to the main menu. Customers may also exit the menu at any time.

The customer menu remains active until the user chooses to return to the main menu (by completing checkout or selecting "Cancel and Return to Main Menu").

4. Parking Administrator Menu

Parking administrators have access to a menu that allows them to:

- View the total profit generated from all parking reservations.
- View the most popular parking duration based on customer selection.
- Return to the main menu

The administrator menu remains active until the user chooses to return to the main menu. he program terminates automatically when all parking slots are occupied. All user input must be validated, and all outputs, including menus and invoices, must be neatly organized using String.format().

Additional Requirements and Guidelines

1. The program must automatically terminate from customer menu when all parking slots are occupied.
2. All user input must be validated.
3. When the user needs to select a parking slot or parking duration, the system must display a menu showing all available options.
4. Code must be clearly organized, properly indented, and commented.
5. At the top of the code, include a comment with the group's students' names, IDs, section number, and Lab Instructor's name.
6. Document all issues encountered during the project along with their solutions. You will be asked to discuss these challenges and explain your design decisions.
7. Only course-covered concepts may be used. Arrays and multiple classes are not allowed (Phase 2 will introduce multiple classes). Only String.format() may be used for formatting, and any use of external material will be graded zero.
8. Plagiarism and the use of AI tools (e.g., ChatGPT, Copilot, Gemini, Claude) are strictly prohibited and will incur severe penalties.

Submission Requirements and Rules

1. **Deadline:** March 31, 2026 at 5:00 PM
2. **Submission** (through LMS):
   - Submit the code as a **.java** file.
   - Submit a copy of the same code as a **.txt** file.